

☐ أ٤

☐ ب٤

☐ ج٤

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6

* رقم الكمبيوتر

17

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

MZQF

 Will be reviewed

8

What is the coordinates of point $p1=(2,2)$ after executing the following OpenGL code segment:

```
glTranslate(2, 0, 0);  
glScalef(1.0, -1.0, 1.0);  
glRotate(180, 0, 0, 1);  
drawPoint_p1(); *
```

☐ (2,2)

☒ (0,2)

☐ (0,-2)

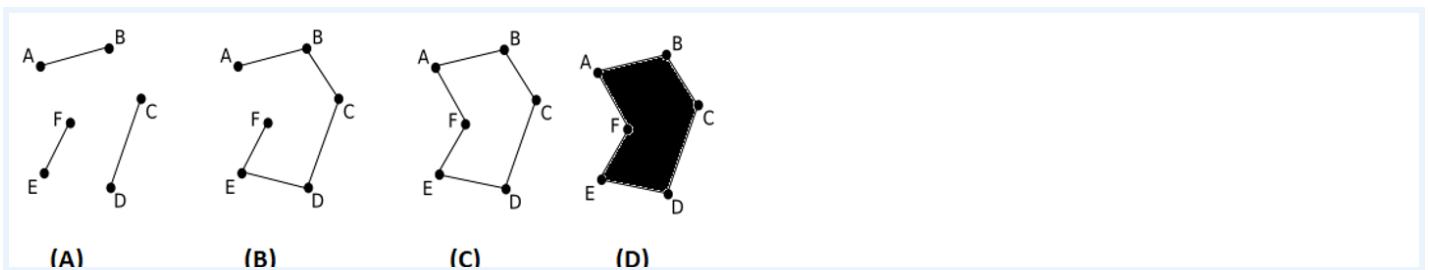
☐ (2,0)

 Will be reviewed

9

The output of executing the following OpenGL code segment is:

```
glBegin(GL_LINE_LOOP);  
glVertex2fv(A);  
glVertex2fv(B);  
glVertex2fv(C);  
glVertex2fv(D);  
glVertex2fv(E);  
glVertex2fv(F);  
glEnd(); *
```



☐ A

☐ B

☒ C

☐ D

 Will be reviewed

10

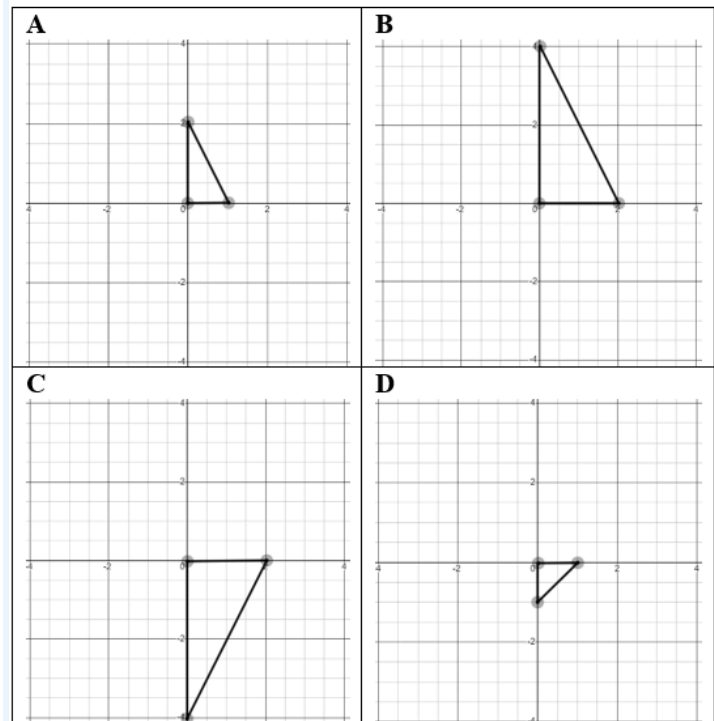
Assume that drawTriangle function draws a triangle (0,0), (1,0), (0,2), what is the output of executing the following code segment?

```
glMatrixMode (GL_PROJECTION);
```

```

glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0);
glMatrixMode(GL_MODELVIEW);
glPushMatrix();
glLoadIdentity();
glScalef(-1.0, -1.0, 1.0);
glRotate(180, 0, 0, 1);
glScalef(2.0, 2.0, 1.0);
drawTriangle();
glPopMatrix(); *

```



☐ A

☒ B

☐ C

☐ D

 Will be reviewed

11

The function `glPolygonMode` allows polygons to be rendered as filled solids, as outlines, or as points. *

☒ True

☐ False

 **Will be reviewed**

12

What is the purpose of the 'LookAt' matrix in the View transformation? *

☐ To project 3D coordinates onto a 2D plane.

☐ To define the size of the window.

☒ To create a coordinate system based on the camera position, a target point, and an up-vector.

☐ To calculate the lighting of the scene.

 **Will be reviewed**

13

To scale an object to be twice as large in all directions, the diagonal elements of the 4x4 matrix should be: *

☐ [0, 0, 0, 1]

☒ [2, 2, 2, 2]

☐ [2, 2, 2, 1]

☐ [1, 1, 1, 2]

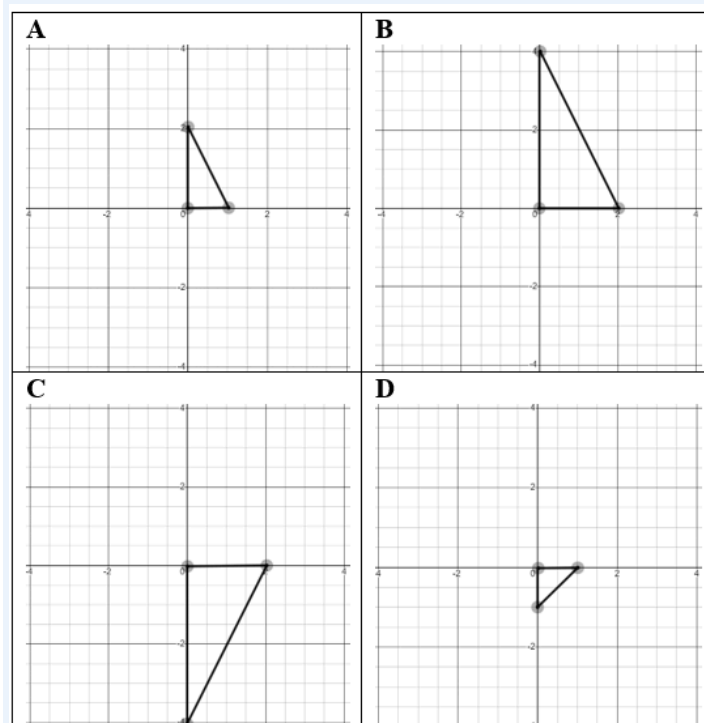
 Will be reviewed

14

Assume that drawTriangle function draws a triangle (0,0), (1,0), (0,2), what is the output of executing the following code segment?

```
glMatrixMode (GL_PROJECTION);  
glLoadIdentity();  
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0);  
glMatrixMode(GL_MODELVIEW);  
glPushMatrix();  
glLoadIdentity();  
glScalef(-1.0, 1.0,1.0);  
glRotate(180, 0, 0, 1);  
glScalef(2.0, -2.0,1.0);  
drawTriangle();  
glPopMatrix();
```

*



☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

15

The RGB of A cyan color is *

☐ (1.0,1.0,0.0)

☐ (1.0,0.0,1.0)

☒ (0.0,1.0,1.0)

☐ (1.0,1.0,1.0)

 **Will be reviewed**

16

What is the primary function of glutMainLoop() in OpenGL? *

☐ Creates a window on the screen.

☒ Enters the main event processing loop.

☐ Renders the scene.

☐ Initializes the OpenGL environment.

 **Will be reviewed**

17

Which GLUT function is used to initialize GLUT? *

- ☒ glutInit()
- ☐ glutMainLoop()
- ☐ glutDisplayFunc()
- ☐ glutCreateWindow()

 **Will be reviewed**

18

The OpenGL command to reflect a 2-D object around the y-axis is glScalef(1,1, 1); *

- ☐ True
- ☒ False

 **Will be reviewed**

19

Which texture filtering method results in a 'blocky' or 'pixelated' look when a texture is scaled up, but is computationally very cheap? *

- ☒ Anisotropic Filtering
- ☐ GL_LINEAR
- ☐ GL_NEAREST
- ☐ GL_MIPMAP

 Will be reviewed

20

Which function is used in OpenGL to set the background color used when clearing the screen? *

- ☐ glBackground()
- ☐ glColor3f()
- ☒ glClearColor()
- ☐ glClear()

 Will be reviewed

21

What type of projection is typically used for 2D games or CAD software where parallel lines remain parallel and objects do not get smaller with distance? *

- ☒ Orthographic Projection
- ☐ Frustum Projection
- ☐ Perspective Projection
- ☐ Mercator Projection

 Will be reviewed

22

OpenGL represents a 4×4 matrix not as a two-dimensional array of floating-point values, but as a single array of 16 floating-point values. *

☒ True

☐ False

 **Will be reviewed**

23

If the scaling factor value is greater than one, then *

☐ It reduces the size of object

☒ It increases the size of object

☐ It does not change the shape of an object

☐ None of the above

 **Will be reviewed**

24

Normalize this vector: (8,-6,0) *

☐ $(\frac{3}{5}, -\frac{4}{5}, 0)$

☒ $(\frac{4}{5}, -\frac{3}{5}, 0)$

☐ $(\frac{4}{5}, \frac{3}{5}, 0)$

☐ (3, 4, 5)

 **Will be reviewed**

25

What kind of argument is expected for glColor3fv command? *

- ☐ Any void variable
- ☒ An array typecasted with (GLfloat *) or in the type indicated by the preceding letter.
- ☐ A GLuint *pointer
- ☐ A virtual method function pointer

 **Will be reviewed**

26

Which component of the graphics pipeline is responsible for storing the final rendered image before it is displayed on the screen? *

- ☒ Z-Buffer
- ☐ Frame Buffer
- ☐ Texture Atlas
- ☐ Vertex Buffer

 **Will be reviewed**

27

Once a 2D rotation is performed, it is not possible to reverse the rotation. *

- ☐ True
- ☒ False

 **Will be reviewed**

28

The main input to the Fragment Processing stage is: *

- ☒ Transformed vertices with attributes.
- ☐ Fragments generated by rasterization.
- ☐ A list of completed 2D triangles.
- ☐ The raw 3D model vertices.

 **Will be reviewed**

29

Which matrix is responsible for moving the viewer in a 3D scene? *

- ☐ Projection Matrix
- ☐ Model Matrix
- ☐ Identity Matrix
- ☒ View Matrix

 **Will be reviewed**

30

What is the standard order of the matrix chain used to transform a vertex v from local space to clip space? *

- ☐ $V' = \text{Model} \times \text{Projection} \times \text{View} \times V$

☒ $V' = \text{Model} \times \text{View} \times \text{Projection} \times V$

☐ $V' = \text{Projection} \times \text{View} \times \text{Model} \times V$

☐ $V' = \text{View} \times \text{Model} \times \text{Projection} \times V$

 **Will be reviewed**

31

If we use the drawing mode `GL_TRIANGLE_STRIP` with six vertices between `glBegin(GL_TRIANGLE_STRIP)` and `glEnd()`, the number of drawn triangles is five. *

☐ True

☒ False

 **Will be reviewed**

32

Which shader stage is responsible for determining the final color of a pixel? *

☐ Tessellation Shader

☐ Vertex Shader

☒ Fragment Shader

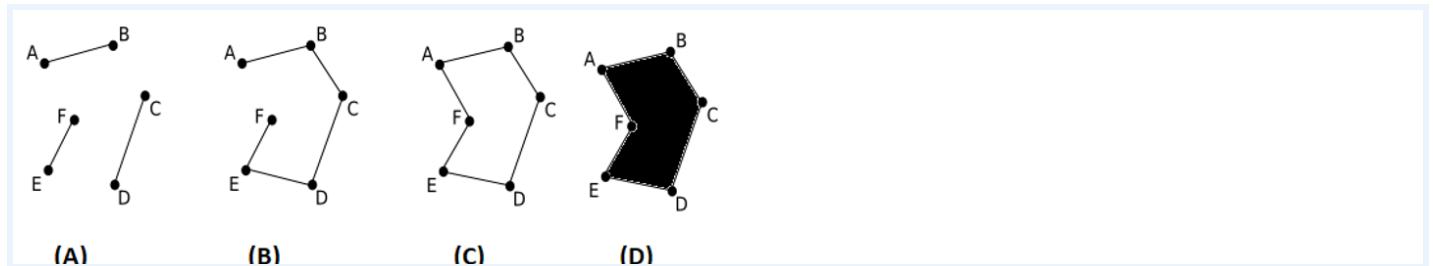
☐ Geometry Shader

 **Will be reviewed**

33

The output of executing the following OpenGL code segment is:
`glBegin(GL_LINE_STRIP);`

```
glVertex2fv(A);
glVertex2fv(B);
glVertex2fv(C);
glVertex2fv(D);
glVertex2fv(E);
glVertex2fv(F);
glEnd(); *
```



- ☐ A
- ☒ B
- ☐ C
- ☐ D

Will be reviewed

34

Which lighting component in the Phong model depends specifically on the position of the viewer? *

- ☐ Emissive
- ☐ Ambient
- ☒ Specular
- ☐ Diffuse

 **Will be reviewed**

35

The graphic pipeline is a series of stages that a 3D model goes through to become a 2D image on a screen. *

☒ True

☐ False

 **Will be reviewed**

36

The position of the vertex after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(1.0f, -1.0f, 0.0f);  
glRotatef(180.0, 0, 0, 1.0);  
glBegin(GL_POINTS);  
glVertex3f(1.0, 0.0, 0.0);  
glEnd(); *
```

☐ (1,-1,0)

☐ (1,0,0)

☒ (0,-1,0)

☐ (-1,0,0)

☐ (-1,-1,0)

 **Will be reviewed**

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(40.0, 0, 0, 1.0);
glRotatef(50.0, 0, 0, 1.0); *
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☒ A

☐ B

☐ C

☐ D

 Will be reviewed

Specifying the projection transformation is like choosing the location for a camera. *

☒ True

☐ False

 Will be reviewed

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
```

```
glScalef(-1.0, 1.0,1.0);  
glScalef(2.0, 1.0, 1.0); *
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B
- ☐ C
- ☒ D

 **Will be reviewed**

40

What is the primary function of glutPostRedisplay in OpenGL? *

- ☐ To enter the main event processing loop.
- ☐ To create a window on the screen.
- ☒ To request a redraw of the current window.
- ☐ To initialize the OpenGL environment.

 **Will be reviewed**

41

Let us scale a square with one unit side length using glScale(2.0f, 0.0f, 0.0f). The output after scaling is: *

- ☐ Square with two units side length.

- ☐ The square will disappear.
- ☒ Rectangle with two units' width and one unit height.
- ☐ Rectangle with two units' height and one unit width.

 **Will be reviewed**

42

Rotating the 3D point $(-1,0,0)$ by -90 around z -axis gives what: *

- ☐ $(1,0,0)$
- ☐ $(-1,0,0)$
- ☒ $(0,1,0)$
- ☐ $(-1,-1,0)$
- ☐ $(0,-1,0)$

 **Will be reviewed**

43

Given three vertices $P1=(1, 1, 0.5)$, $P2=(1.5, 0.5, 0.5)$, $P3=(1, 0.5, 1)$ on a plane, the equation of the plane is *

- ☐ $x+2y+2z=3.5$
- ☐ $x+2y+z=3$
- ☒ $x+y+z=2.5$
- ☐ $x+y+0.5z=2.25$

 **Will be reviewed**

44

Which of the following is a 'Primitive' in OpenGL? *

☒ Triangles

☐ Cubes

☐ Squares

☐ Circles

 **Will be reviewed**

45

In glVertex2i() command, 2 represents _____ *

☐ None of these

☒ number of arguments

☐ type of arguments

☐ basic command

 **Will be reviewed**

46

If you want to render a scene to a specific portion of a window, you would use: *

☐ glFrustum()

☐ gluPerspective()

☒ glViewport()

☐ glOrtho()

📝 Will be reviewed

47

What is the result of the matrix-vector multiplication $M \times V$ where M is a Translation matrix, and V has a w component of 0? *

☐ The vector remains unchanged.

☐ The vector is scaled to zero.

☐ The calculation results in an error.

☒ The vector is translated by the offsets in M .

📝 Will be reviewed

48

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(-1.0f, -2.0f, -3.0f);  
glRotatef(90.0, 0, 0, 1.0);  
glPushMatrix();  
glTranslatef(-1.0f, -2.0f, -3.0f);  
glRotatef(-90.0, 0, 0, 1.0);  
glScalef (-1.0,0.5, -1.0);  
glPopMatrix();  
glRotatef(90.0, 0, 0, 1.0);  
glTranslatef(-1.0f, -2.0f, 3.0f); *
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 2 \\ -1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☒ C

☐ D

 Will be reviewed

49

The Modelview matrix after executing the following OpenGL code segment is:

`glMatrixMode(GL_MODELVIEW);`

`glLoadIdentity();`

`glRotatef(90.0, 0, 0, 1.0); *`

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☒ A

☐ B

☐ C

☐ D

 Will be reviewed

50

The primary function of the Vertex Processor is to: *

- ☐ Store the final rendered image.
- ☒ Manipulate the geometric data of a 3D model.
- ☐ Convert 3D shapes into 2D fragments.
- ☐ Determine the final color of each pixel.

📝 Will be reviewed

51

What does the 'Z-buffer' (Depth Buffer) prevent? *

- ☐ Texture distortion
- ☐ Low frame rates
- ☐ Program crashes
- ☒ Objects in the background from being drawn over objects in the foreground.

📝 Will be reviewed

52

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef(-1.0, 2.0, -1.0);
glRotatef(90.0, 0, 0, 1.0); *
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B

☐ C

☒ D

 **Will be reviewed**

53

Which axis remains unchanged when applying the following rotation matrix: *

$$\begin{bmatrix} 0 & -3 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

☐ The Y-axis

☐ The X-axis

☒ The Z-axis

☐ The W-axis

 **Will be reviewed**

54

Given a standard 4x4 column-major translation matrix, where are the translation offsets tx, ty, and tz located? *

☒ In the last column.

☐ Along the main diagonal.

☐ They are not stored in the matrix.

☐ In the top-left 2x2 sub-matrix.

 **Will be reviewed**

55

Rotate (z, 90) * Rotate (z, 45) is equal to Rotate (z, 45) * Rotate (z, 90) *

☒ True

☐ False

 **Will be reviewed**

56

OpenGL can draw convex polygons. *

☒ True

☐ False

 **Will be reviewed**

57

Which type of projection mimics the human eye, where objects further away appear smaller? *

☐ Oblique Projection

☐ Orthographic Projection

☒ Perspective Projection



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